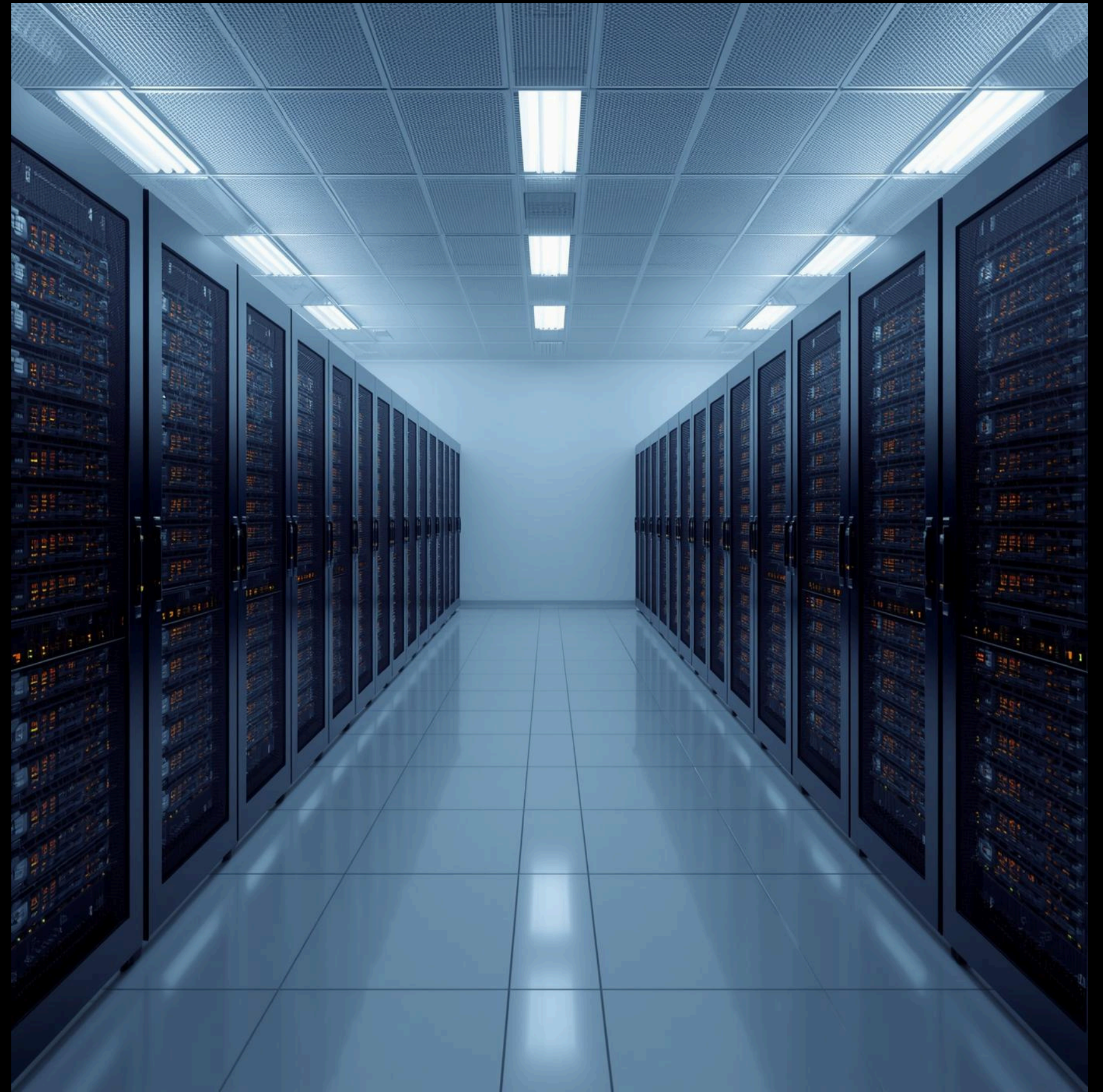


PROJECT PRESENTATION

Database Systems Project Presentation

Presented by: Group Oracle



Team Introduction



Ali Huseynov

Fronted / Backend Operations and
UI Views
Countries Table



Devrim Polat

Backend Operations,
Economy Indicators Table



Yunus Korkmaz

Database Management,
Health Indicators Table



Ahmet Yusuf Kurukiz

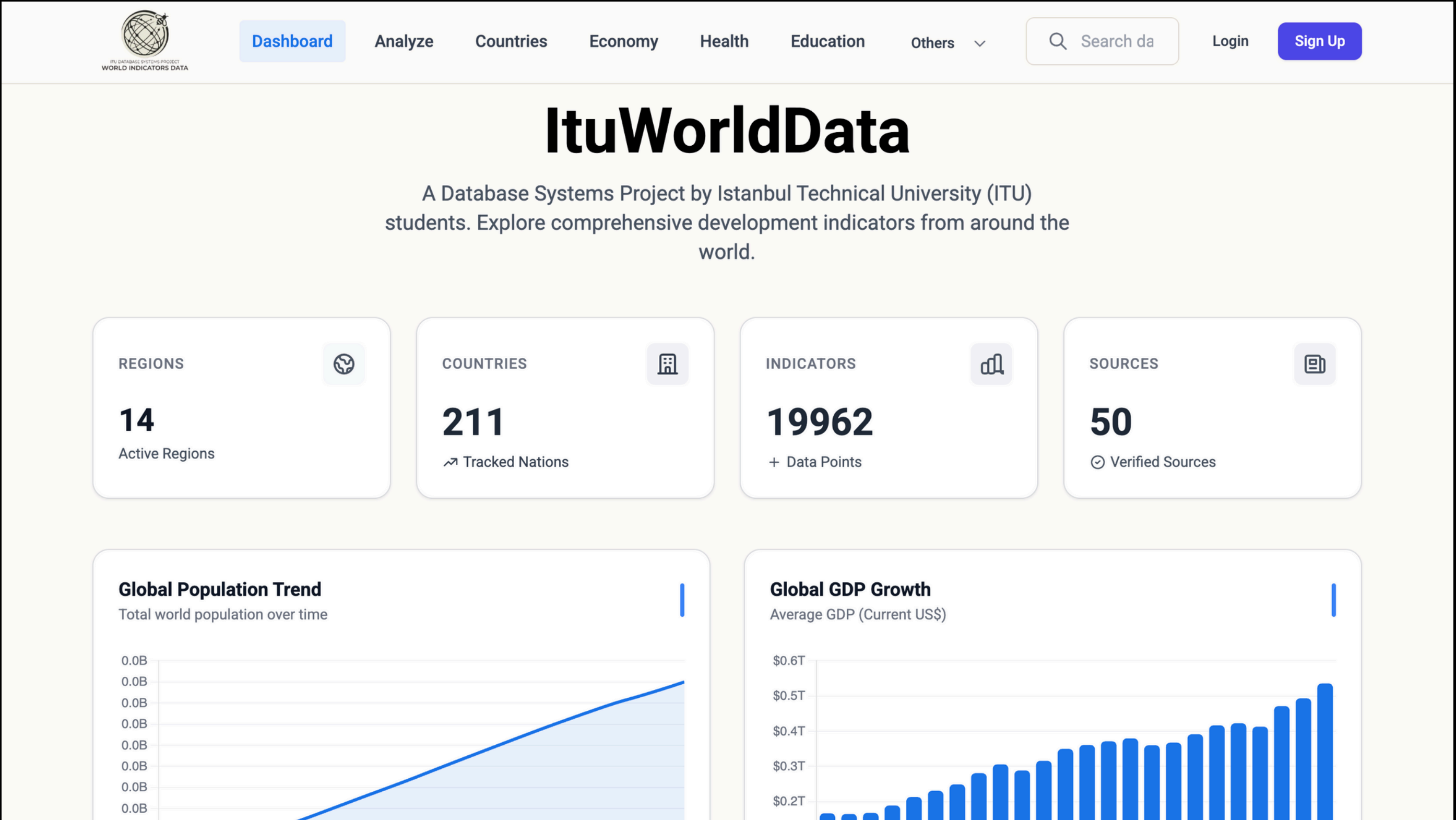
Backend Operations
Indicators Table



Mehmet Fatih Kaya

Frontend Operations and UI Views,
Education Indicators Table

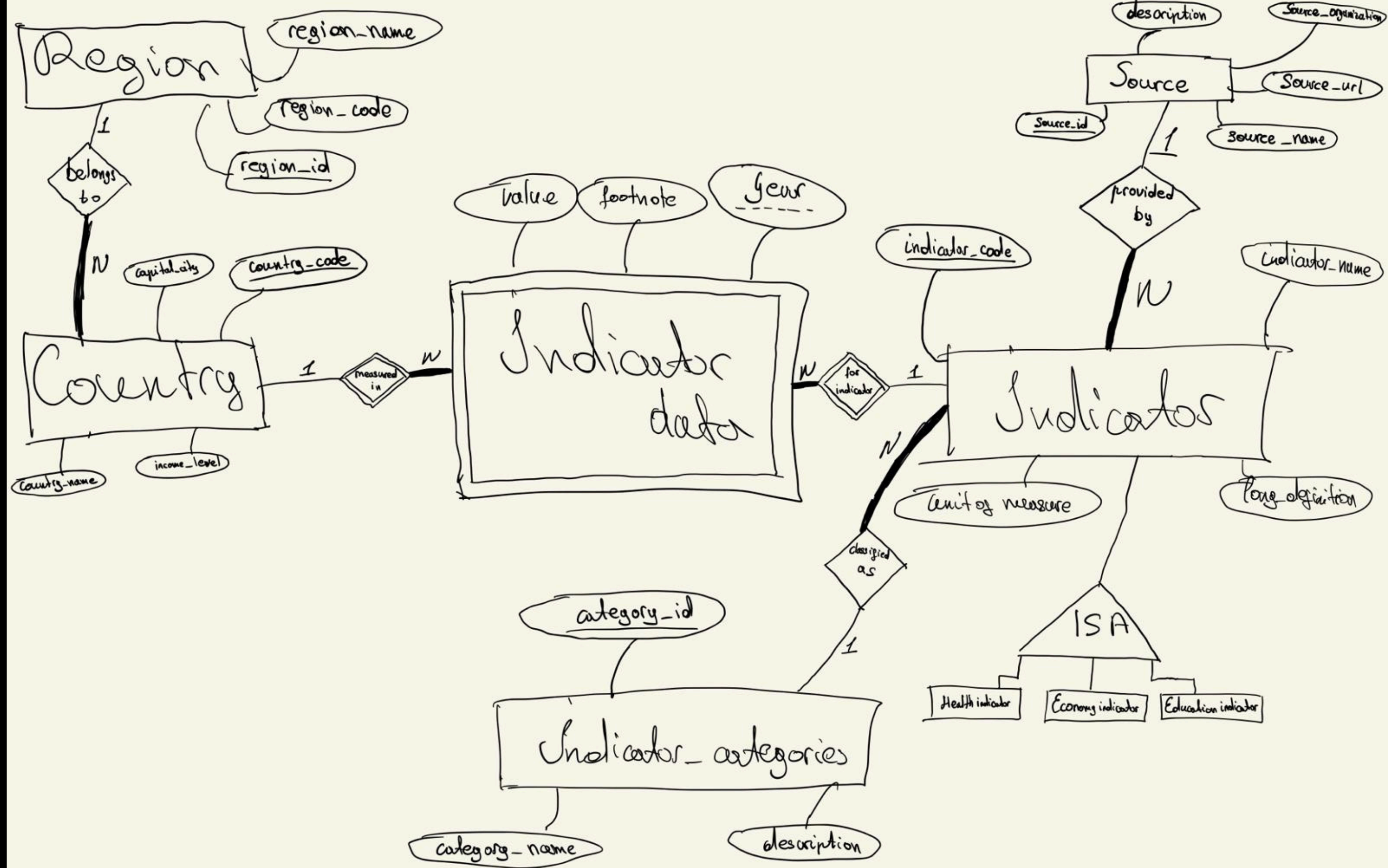
Overview



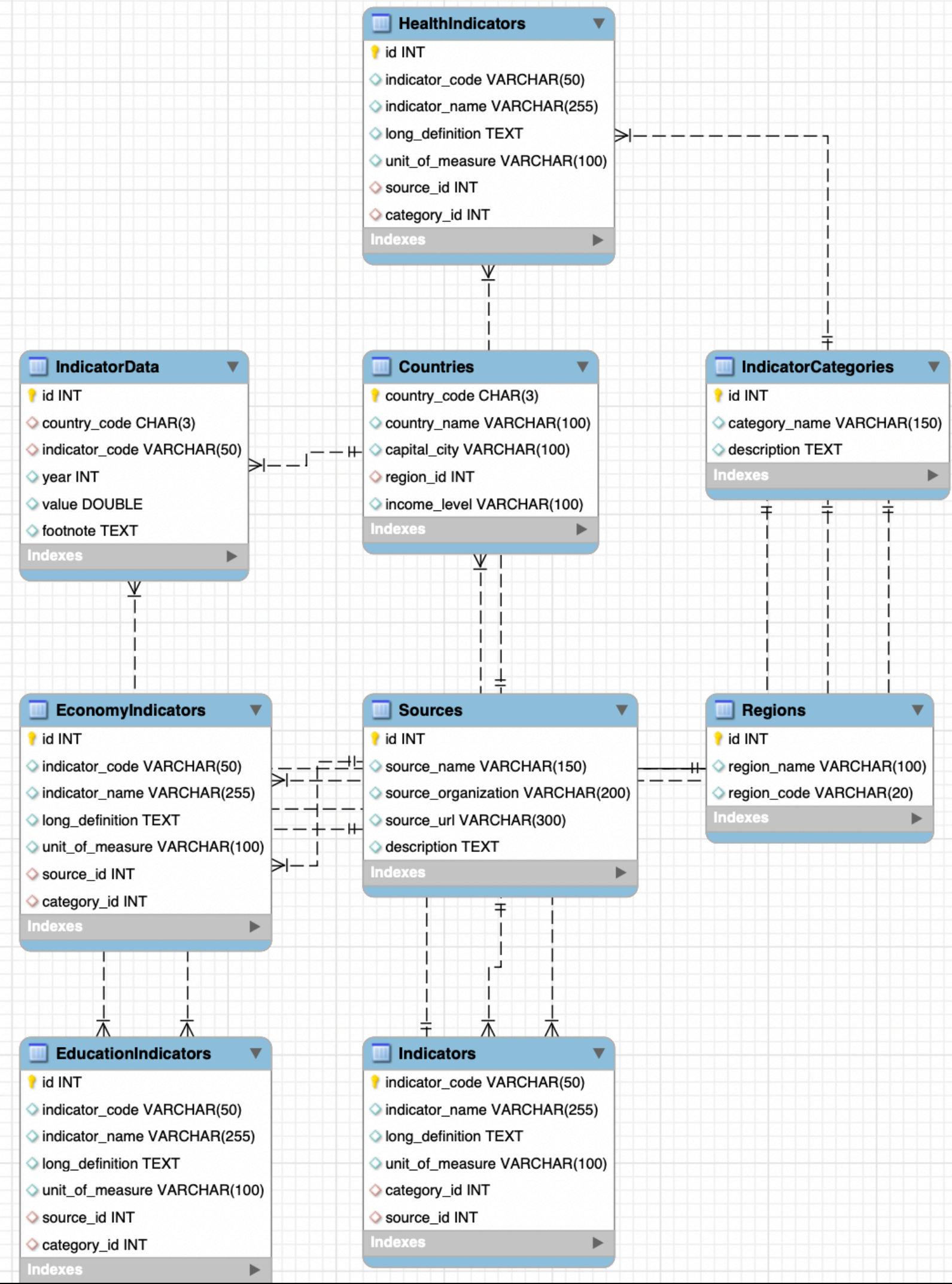
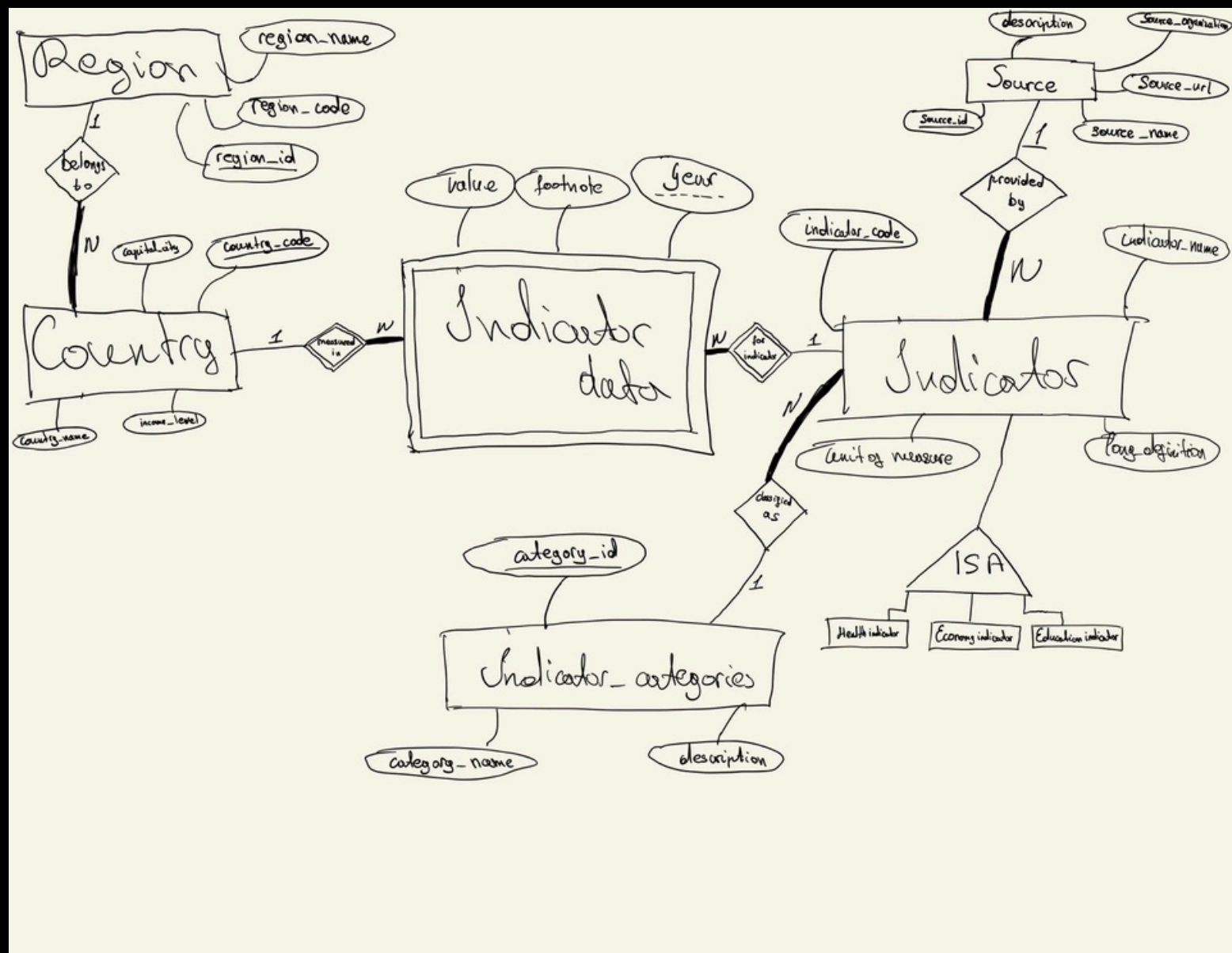
Motivation and Goals

- **Data Fragmentation:** Critical global data is scattered across disparate sources (World Bank, WHO, UNESCO), making cross-domain analysis impossible
- **Redundancy:** Traditional flat-file storage leads to massive data duplication and update anomalies.
- **Normalization:** To design a 3NF schema that ensures data consistency
- **Security:** To implement a "Strict Admin Approval System" where new users must be explicitly approved by an Admin before accessing the data
- **Insight:** To transform raw static data into dynamic, comparative time-series visualizations

ER Diagram



ER-to-Relational Mapping:



Before Normalization

Before Normalization: Raw JSON

- Input: Raw API response in nested format.
- Redundancy: Source details are duplicated in thousands of rows.
- Inefficiency: High storage cost and risk of update anomalies.

```
t was reported by the donor to the Global Partnership for Education.,"sourceOrganization":"The 2011 Monitoring Exercise on Aid Effectiveness i
Global Partnership for Education in parallel to the OECD's 2011 Survey on Monitoring the Paris Declaration. Through this exercise parti
llocated and projected to specific developing countries. Data were later validated and/or updated when preparing the GPE Results Report
011 and in USD. When it was not the case, currencies were converted to USD. Data are relative to a calendar year, unless it is specified
in this 2012 Monitoring Exercise, please refer to the following web site: http://www.globalpartnership.org/our-work/areas-of-focus/aid-effectiveness-2/","topics":[]},{id":"5.1.1_NER.TOTA.AID.AFD","name":"International aid disbursed to total education, AFD to Niger (US$100 million)"},"34","value":"Global Partnership for Education"},"sourceNote":"International concessional aid disbursed by the reporting development partner to the specific developing country. Targets indicate scheduled or projected aid. Accounted aid includes activities, projects, technical cooperation reported by the donor to the Global Partnership for Education.,"sourceOrganization":"The 2011 Monitoring Exercise on Aid Effectiveness i
Partnership for Education in parallel to the OECD's 2011 Survey on Monitoring the Paris Declaration. Through this exercise participatin
d and projected to specific developing countries. Data were later validated and/or updated when preparing the GPE Results Report in 2011
in USD. When it was not the case, currencies were converted to USD. Data are relative to a calendar year, unless it is specified otherwi
s 2012 Monitoring Exercise, please refer to the following web site: http://www.globalpartnership.org/our-work/areas-of-focus/aid-effectiveness-2/","topics":[]},{id":"5.1.1_RWA.TOTA.AID.DFID","name":"International aid disbursed to total education, DFID to Rwanda (US$100 million)"},"34","value":"Global Partnership for Education"},"sourceNote":"International concessional aid disbursed by the reporting development partner to the specific developing country. Targets indicate scheduled or projected aid. Accounted aid includes activities, projects, technical cooperation reported by the donor to the Global Partnership for Education.,"sourceOrganization":"The 2011 Monitoring Exercise on Aid Effectiveness i
Partnership for Education in parallel to the OECD's 2011 Survey on Monitoring the Paris Declaration. Through this exercise participatin
d and projected to specific developing countries. Data were later validated and/or updated when preparing the GPE Results Report in 2011
in USD. When it was not the case, currencies were converted to USD. Data are relative to a calendar year, unless it is specified otherwi
s 2012 Monitoring Exercise, please refer to the following web site: http://www.globalpartnership.org/our-work/areas-of-focus/aid-effectiveness-2/","topics":[]},{id":"5.1.1_SEN.TOTA.AID.CIDA","name":"International aid disbursed to total education, CIDA to Senegal (US$100 million)"},"34","value":"Global Partnership for Education"},"sourceNote":"International concessional aid disbursed by the reporting development partner to the specific developing country. Targets indicate scheduled or projected aid. Accounted aid includes activities, projects, technical cooperation reported by the donor to the Global Partnership for Education.,"sourceOrganization":"The 2011 Monitoring Exercise on Aid Effectiveness i
Partnership for Education in parallel to the OECD's 2011 Survey on Monitoring the Paris Declaration. Through this exercise participatin
d and projected to specific developing countries. Data were later validated and/or updated when preparing the GPE Results Report in 2011
in USD. When it was not the case, currencies were converted to USD. Data are relative to a calendar year, unless it is specified otherwi
s 2012 Monitoring Exercise, please refer to the following web site: http://www.globalpartnership.org/our-work/areas-of-focus/aid-effectiveness-2/","topics":[]},{id":"5.1.1_SLE.TOTA.AID.DFID","name":"International aid disbursed to total education, DFID to Sierra Leone (US$100 million)"},"34","value":"Global Partnership for Education"},"sourceNote":"International concessional aid disbursed by the reporting development partner to the specific developing country. Targets indicate scheduled or projected aid. Accounted aid includes activities, projects, technical cooperation reported by the donor to the Global Partnership for Education.,"sourceOrganization":"The 2011 Monitoring Exercise on Aid Effectiveness i
Partnership for Education in parallel to the OECD's 2011 Survey on Monitoring the Paris Declaration. Through this exercise participatin
d and projected to specific developing countries. Data were later validated and/or updated when preparing the GPE Results Report in 2011
in USD. When it was not the case, currencies were converted to USD. Data are relative to a calendar year, unless it is specified otherwi
s 2012 Monitoring Exercise, please refer to the following web site: http://www.globalpartnership.org/our-work/areas-of-focus/aid-effectiveness-2/","topics":[]},{id":"5.1.1_VNM.TOTA.AID.BEL","name":"International aid disbursed to total education, Belgium to Vietnam (US$100 million)"},"34","value":"Global Partnership for Education"},"sourceNote":"International concessional aid disbursed by the reporting development partner to the specific developing country. Targets indicate scheduled or projected aid. Accounted aid includes activities, projects, technical cooperation reported by the donor to the Global Partnership for Education.,"sourceOrganization":"The 2011 Monitoring Exercise on Aid Effectiveness i
Partnership for Education in parallel to the OECD's 2011 Survey on Monitoring the Paris Declaration. Through this exercise participatin
d and projected to specific developing countries. Data were later validated and/or updated when preparing the GPE Results Report in 2011
in USD. When it was not the case, currencies were converted to USD. Data are relative to a calendar year, unless it is specified otherwi
s 2012 Monitoring Exercise, please refer to the following web site: http://www.globalpartnership.org/our-work/areas-of-focus/aid-effectiveness-2/","topics":[]},{id":"5.1.1_ZMB.TOTA.AID.DNK","name":"International aid disbursed to total education, Denmark to Zambia (US$100 million)"},"34","value":"Global Partnership for Education"},"sourceNote":"International concessional aid disbursed by the reporting development partner to the specific developing country. Targets indicate scheduled or projected aid. Accounted aid includes activities, projects, technical cooperation reported by the donor to the Global Partnership for Education.,"sourceOrganization":"The 2011 Monitoring Exercise on Aid Effectiveness i
Global Partnership for Education in parallel to the OECD's 2011 Survey on Monitoring the Paris Declaration. Through this exercise participatin
llocated and projected to specific developing countries. Data were later validated and/or updated when preparing the GPE Results Report
011 and in USD. When it was not the case, currencies were converted to USD. Data are relative to a calendar year, unless it is specified
in this 2012 Monitoring Exercise, please refer to the following web site: http://www.globalpartnership.org/our-work/areas-of-focus/aid-effectiveness-2/","topics":[]},{id":"5.1.10_AFG.TOTA.AID.SIDA","name":"International aid disbursed to total education, Sida to Afghanistan (US$100 million)"},"34","value":"Global Partnership for Education"},"sourceNote":"International concessional aid disbursed by the reporting development partner to the specific developing country. Targets indicate scheduled or projected aid. Accounted aid includes activities, projects, technical cooperation reported by the donor to the Global Partnership for Education.,"sourceOrganization":"The 2011 Monitoring Exercise on Aid Effectiveness i
Global Partnership for Education in parallel to the OECD's 2011 Survey on Monitoring the Paris Declaration. Through this exercise participatin
llocated and projected to specific developing countries. Data were later validated and/or updated when preparing the GPE Results Report
011 and in USD. When it was not the case, currencies were converted to USD. Data are relative to a calendar year, unless it is specified
```


Normalization

- ETL Approach: We used a Python script to Extract, Transform, and Load the data directly in 3NF structure.
- In-Memory De-duplication:
 - Utilized Python dictionaries (source_map) to identify and filter unique entries before database insertion.
 - Prevents duplicate records in Sources and Categories tables.
- Foreign Key Association:
 - The script retrieves the unique ID of the source/category.
 - Inserts only this Foreign Key (sid) into the main Indicators table, ensuring data integrity.

```
111     # source
112     src = ind.get("source", {})
113     sid = src.get("id")
114
115     cursor.execute(sql, (
116         code,
117         name,
118         long_def,
119         unit,
120         tid,
121         sid
122     ))
```

```
144
145     # Collect sources
146     source_map = {}
147     for ind in indicators:
148         src = ind.get("source", {})
149         sid = src.get("id")
150         if sid:
151             source_map[sid] = {
152                 "name": src.get("value"),
153                 "organization": src.get("organization"),
154                 "url": src.get("url")
155             }
156     print(f"Unique sources: {len(source_map)}")
157     insert_sources(source_map)
```


After Normalization

- Structured Data: The complex, nested JSON objects are now stored in relational tables (Indicators, Sources, Categories).
- Reference-Based Storage: The "Source" column is no longer a repeated text block. It is a Foreign Key reference to the Sources table.
- 3NF Compliance:
- Transitive Dependencies Removed: Source details (organization name, description) are separated from the Indicator data.
- Update Anomalies Eliminated: Updating a source's name in the Sources table automatically updates it for all associated indicators.

 Dashboard Analyze Countries Economy Health Education Others

Welcome, ali [Logout](#)

All Indicators

Browse and manage all available indicators

CODE	NAME	CATEGORY	SOURCE
8.3.2_CAF.BAC.SUCC	Baccalaureate in Central African Republic, exam at the end of secondary education, success rate (%)	None	Global Partnership for Education
8.3.1_CAF.BREVET.SUCC	Brevet des colleges in Central African Republic, success rate (%)	None	Global Partnership for Education
5.2.1_LAO.BAS.AID.ADB	International aid disbursed to basic education, ADB to Laos (USD million)	None	Global Partnership for Education
5.2.2_SEN.BAS.AID.FR	International aid disbursed to basic education, AFD and French Embassy to Senegal (USD million)	None	Global Partnership for Education
5.2.2_BFA.BAS.AID.AFD	International aid disbursed to basic education, AFD to Burkina Faso (USD million)	None	Global Partnership for Education
5.2.1_KHM.BAS.AID.BAD	International aid disbursed to basic education, AfDB to Cambodia (USD million)	None	Global Partnership for Education

Dataset Description

What does the data include?: The dataset comprises time-series development metrics categorized into three core domains: Economy, Health, and Education. It covers global data for countries and regions, including variables such as GDP, Population, Life Expectancy, and School Enrollment rates.

What is the source?: The primary source is the World Development Indicators (WDI) database managed by the World Bank. We utilized the official WDI API and bulk CSV exports to ensure data accuracy and reliability.

How large is it?: The database contains 19962 rows, covering information about 211 countries.

Data Fetching

```
def fetch_indicator_all(indicator_code):
    url = (
        f"https://api.worldbank.org/v2/country/all/indicator/"
        f"{indicator_code}?format=json&per_page=20000"
    )

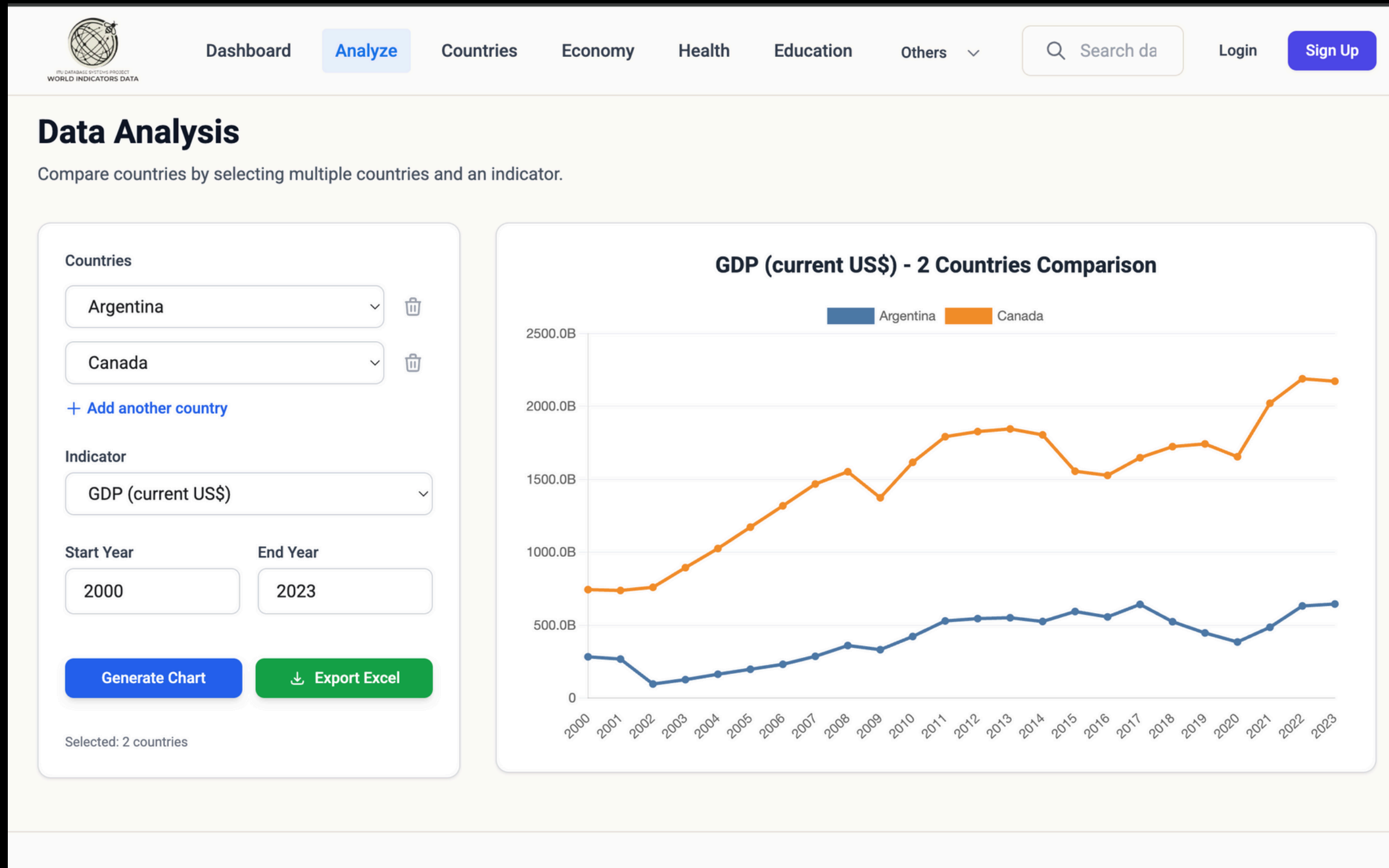
    for attempt in range(5):
        resp = requests.get(url)
        if resp.status_code == 200:
            data = resp.json()
            if len(data) < 2 or data[1] is None:
                return []
            return data[1]

        if resp.status_code in (500, 502, 503, 504):
            print(f"[{indicator_code}] Server {resp.status_code}, retry {attempt+1}/5")
            time.sleep(1)
            continue

        resp.raise_for_status()

    print(f"[{indicator_code}] SKIPPED: API unreachable.")
    return []
```

Compare Countries



Sign Up and Log In



Create Account

Sign up to access admin features

Username

Email Address

Password

Confirm Password

Sign Up

Already have an account? [Login](#)



Admin Login

Please sign in to access the admin panel

Username

Password

Sign In

```
4 def init_db():
5     conn = get_db_connection()
6     cursor = conn.cursor()
7
8     # Create Users table
9     with open('database/create_users_table.sql', 'r') as f:
10         cursor.execute(f.read())
11
12     # Check if admin exists
13     cursor.execute("SELECT * FROM Users WHERE username = 'huseynov'")
14     if not cursor.fetchone():
15         # Create admin user
16         password = "09052006"
17         password_hash = generate_password_hash(password, method='pbkdf2:sha256')
18         cursor.execute(
19             "INSERT INTO Users (username, password_hash, role) VALUES (%s, %s, %s)",
20             ('huseynov', password_hash, 'admin')
21         )
22         print("Admin user 'huseynov' created.")
23     else:
24         print("Admin user already exists.")
25
26     conn.commit()
27     cursor.close()
28     conn.close()
29
30 if __name__ == '__main__':
31     init_db()
```


Complex Queries

```
def get_education_indicators(page=1, per_page=1000, search_query=None):
    offset = (page - 1) * per_page

    if search_query:
        count_query = """
        SELECT COUNT(*) as count
        FROM EducationIndicators i
        WHERE i.indicator_name LIKE %s OR i.indicator_code LIKE %s
        """
        count_result = execute(count_query, (f"%{search_query}%", f"%{search_query}%"), fetch=True)
        total_count = count_result[0]['count'] if count_result else 0

        items = execute(f"""
        SELECT i.*, s.source_name, c.category_name,
        EXISTS(SELECT 1 FROM IndicatorData d WHERE d.indicator_code = i.indicator_code) as has_data,
        (CASE WHEN i.long_definition IS NOT NULL AND i.long_definition != '' THEN 1 ELSE 0 END) as is_well_defined
        FROM EducationIndicators i
        LEFT JOIN Sources s ON i.source_id = s.id
        LEFT JOIN IndicatorCategories c ON i.category_id = c.id
        WHERE i.indicator_name LIKE %s OR i.indicator_code LIKE %s
        ORDER BY has_data DESC, is_well_defined DESC, i.indicator_name ASC
        LIMIT {per_page} OFFSET {offset}
        """, (f"%{search_query}%", f"%{search_query}%"), fetch=True)
    else:
        count_result = execute("SELECT COUNT(*) as count FROM EducationIndicators", fetch=True)
        total_count = count_result[0]['count'] if count_result else 0

        items = execute(f"""
        SELECT i.*, s.source_name, c.category_name,
        EXISTS(SELECT 1 FROM IndicatorData d WHERE d.indicator_code = i.indicator_code) as has_data,
        (CASE WHEN i.long_definition IS NOT NULL AND i.long_definition != '' THEN 1 ELSE 0 END) as is_well_defined
        FROM EducationIndicators i
        LEFT JOIN Sources s ON i.source_id = s.id
        LEFT JOIN IndicatorCategories c ON i.category_id = c.id
        ORDER BY has_data DESC, is_well_defined DESC, i.indicator_name ASC
        LIMIT {per_page} OFFSET {offset}
        """, fetch=True)

    return items, total_count
```

127.0.0.1:5001/education?search=school

Dashboard

Analyze

Countries

Economy

Health

Education

Others

Search da

Login

Sign Up

Education Indicators

Explore key education metrics and statistics

Back to All

school

INDICATOR NAME	CATEGORY	SOURCE	ACTIONS
School enrollment, primary (% gross) SE.PRM.ENRR Data Available	Education	Unknown	View Data
Adequacy of benefits (%) - School Feeding per_sa_sf.adq_pop_tot	Social Protection & Labor	The Atlas of Social Protection: Indicators of Resilience and Equity	View Data
Adequacy of benefits (%) - School Feeding -rural per_sa_sf.adq_pop_rur	Social Protection & Labor	The Atlas of Social Protection: Indicators of Resilience and Equity	View Data
Adequacy of benefits (%) - School Feeding -urban per_sa_sf.adq_pop_urb	Social Protection & Labor	The Atlas of Social Protection: Indicators of Resilience and Equity	View Data
Adequacy of benefits in 1st quintile (poorest) (%) - School feeding per_sa_sf.adq_q1_tot	Social Protection & Labor	The Atlas of Social Protection: Indicators of Resilience and Equity	View Data
Adequacy of benefits in 1st quintile (poorest) (%) - School feeding -rural per_sa_sf.adq_q1_rur	Social Protection & Labor	The Atlas of Social Protection: Indicators of Resilience and Equity	View Data
Adequacy of benefits in 1st quintile (poorest) (%) - School feeding -urban per_sa_sf.adq_q1_urb	Social Protection & Labor	The Atlas of Social Protection: Indicators of Resilience and Equity	View Data
Adequacy of benefits in 1st quintile (poorest) (%) - School feeding (preT) per_sa_sf.adq_q1_preT_tot	Social Protection & Labor	The Atlas of Social Protection: Indicators of Resilience and Equity	View Data
Adequacy of benefits in 2nd quintile (%) - School Feeding per_sa_sf.adq_q2_tot	Social Protection & Labor	The Atlas of Social Protection: Indicators of Resilience and Equity	View Data

Complex Queries

```
def get_health_indicators(page=1, per_page=1000, search_query=None):
    offset = (page - 1) * per_page

    if search_query:
        count_query = """
            SELECT COUNT(*) as count
            FROM HealthIndicators i
            WHERE i.indicator_name LIKE %s OR i.indicator_code LIKE %s
        """
        count_result = execute(count_query, (f"%{search_query}%", f"%{search_query}%"), fetch=True)
        total_count = count_result[0]['count'] if count_result else 0

        items = execute(f"""
            SELECT i.*, s.source_name, c.category_name,
            EXISTS(SELECT 1 FROM IndicatorData d WHERE d.indicator_code = i.indicator_code) as has_data
            FROM HealthIndicators i
            LEFT JOIN Sources s ON i.source_id = s.id
            LEFT JOIN IndicatorCategories c ON i.category_id = c.id
            WHERE i.indicator_name LIKE %s OR i.indicator_code LIKE %s
            ORDER BY has_data DESC, i.indicator_name ASC
            LIMIT {per_page} OFFSET {offset}
        """, (f"%{search_query}%", f"%{search_query}%"), fetch=True)
    else:
        count_result = execute("SELECT COUNT(*) as count FROM HealthIndicators", fetch=True)
        total_count = count_result[0]['count'] if count_result else 0

        items = execute(f"""
            SELECT i.*, s.source_name, c.category_name,
            EXISTS(SELECT 1 FROM IndicatorData d WHERE d.indicator_code = i.indicator_code) as has_data
            FROM HealthIndicators i
            LEFT JOIN Sources s ON i.source_id = s.id
            LEFT JOIN IndicatorCategories c ON i.category_id = c.id
            ORDER BY has_data DESC, i.indicator_name ASC
            LIMIT {per_page} OFFSET {offset}
        """, fetch=True)

    return items, total_count
```

127.0.0.1:5001/health?search=Life

DashboardAnalyzeCountriesEconomyHealthEducationOthers

Search da

LoginSign Up

Health Indicators

Explore key health metrics and statistics

Back to All

Life

INDICATOR NAME	CATEGORY	SOURCE	ACTIONS
Life expectancy at birth, total (years) SP.DYN.LE00.IN Data Available	Uncategorized	Unknown	View Data

Complex Queries

```
def get_economy_indicators(page=1, per_page=1000, search_query=None):
    offset = (page - 1) * per_page

    if search_query:
        count_query = """
            SELECT COUNT(*) as count
            FROM EconomyIndicators i
            WHERE i.indicator_name LIKE %s OR i.indicator_code LIKE %s
            """
        count_result = execute(count_query, (f"%{search_query}%", f"%{search_query}%"), fetch=True)
        total_count = count_result[0]['count'] if count_result else 0

        items = execute(f"""
            SELECT i.*, s.source_name, c.category_name,
            (SELECT COUNT(*) FROM IndicatorData d WHERE d.indicator_code = i.indicator_code) as has_data
            FROM EconomyIndicators i
            LEFT JOIN Sources s ON i.source_id = s.id
            LEFT JOIN IndicatorCategories c ON i.category_id = c.id
            WHERE i.indicator_name LIKE %s OR i.indicator_code LIKE %s
            ORDER BY has_data DESC, i.indicator_name ASC
            LIMIT {per_page} OFFSET {offset}
            """, (f"%{search_query}%", f"%{search_query}%"), fetch=True)
    else:
        count_result = execute("SELECT COUNT(*) as count FROM EconomyIndicators", fetch=True)
        total_count = count_result[0]['count'] if count_result else 0

        items = execute(f"""
            SELECT i.*, s.source_name, c.category_name,
            (SELECT COUNT(*) FROM IndicatorData d WHERE d.indicator_code = i.indicator_code) as has_data
            FROM EconomyIndicators i
            LEFT JOIN Sources s ON i.source_id = s.id
            LEFT JOIN IndicatorCategories c ON i.category_id = c.id
            ORDER BY has_data DESC, i.indicator_name ASC
            LIMIT {per_page} OFFSET {offset}
            """, fetch=True)

    return items, total_count
```

The screenshot shows a web browser at the address 127.0.0.1:5001/economy?search=gdp. The page has a navigation bar with links for Dashboard, Analyze, Countries, Economy (active), Health, Education, and Others. A search bar contains 'gdp'. The main heading is 'Economy Indicators' with a subtitle 'Explore key economic metrics and statistics'. A 'Back to All' button is in the top right. Below the search bar is a table of indicators. Each row includes an indicator name with a code and a 'Data Available' status, a category, a source, and a 'View Data' link.

INDICATOR NAME	CATEGORY	SOURCE	ACTIONS
GDP (current US\$) NY.GDP.MKTP.CD Data Available	Economy & Growth	World Development Indicators	View Data
GDP per capita (current US\$) NY.GDP.PCAP.CD Data Available	Economy & Growth	World Development Indicators	View Data
Trade (% of GDP) NE.TRD.GNFS.ZS Data Available	Economy & Growth	World Development Indicators	View Data
2005 PPP conversion factor, GDP (LCU per international \$) PA.NUS.PPP.05	Economy & Growth	WDI Database Archives	View Data
Agricultural support estimate (% of GDP) NY.AGR.SUBS.GD.ZS	Millenium development goals	Millennium Development Goals	View Data
Agriculture, forestry, and fishing, value added (% of GDP) NV.AGR.TOTL.ZS	Agriculture & Rural Development	World Development Indicators	View Data
Agriculture: contribution to growth of GDP (%) NP.AGR.TOTL.ZG	Economy & Growth	WDI Database Archives	View Data
Annual investment needs for coastal protection, by risk strategy (% of GDP) - constant relative flood risk CC.INCP.KRGC	Economy & Growth	Country Climate and Development Report (CCDR)	View Data
Annual investment needs for coastal protection, by risk strategy (% of GDP) - low risk tolerance CC.INCP.ALRS	Economy & Growth	Country Climate and Development Report (CCDR)	View Data
Annual investment needs for coastal protection, by risk strategy (% of GDP) - optimal protection	Economy & Growth	Country Climate and Development Report (CCDR)	View Data

Complex Queries

```
def get_indicators_with_data():  
    """Get only indicators that have actual data in IndicatorData table"""  
    return execute(  
        """  
        SELECT DISTINCT I.indicator_code, I.indicator_name, I.source_id, I.category_id, I.long_definition,  
        C.category_name, S.source_name  
        FROM Indicators I  
        INNER JOIN IndicatorData D ON I.indicator_code = D.indicator_code  
        LEFT JOIN IndicatorCategories C ON I.category_id = C.id  
        LEFT JOIN Sources S ON I.source_id = S.id  
        ORDER BY I.indicator_name  
        """,  
        fetch=True  
    )
```

Complex Queries

```
def get_yearly_indicator_average(indicator_code):  
    return execute(  
        """  
        SELECT year, AVG(value) AS avg_value  
        FROM IndicatorData  
        WHERE indicator_code=%s  
        GROUP BY year  
        ORDER BY year  
        """,  
        (indicator_code,),  
        fetch=True  
    )  
  
def get_global_population():  
    return get_yearly_indicator_average("SP.POP.TOTL")
```


Complex Queries

```
def get_yearly_indicator_average(indicator_code):  
    return execute(  
        """  
        SELECT year, AVG(value) AS avg_value  
        FROM IndicatorData  
        WHERE indicator_code=%s  
        GROUP BY year  
        ORDER BY year  
        """,  
        (indicator_code,),  
        fetch=True  
    )  
  
def get_global_population():  
    return get_yearly_indicator_average("SP.POP.TOTL")
```

Delete Constraints

- **IndicatorData Table :**
 - **Countries:** ON DELETE RESTRICT
 - **Indicators:** ON DELETE RESTRICT
- **EducationIndicators Table :**
 - **IndicatorCategories:** ON DELETE CASCADE
 - **Sources:** ON DELETE RESTRICT
- **EconomyIndicators Table :**
 - **IndicatorCategories:** ON DELETE RESTRICT
 - **Sources:** ON DELETE RESTRICT
- **Health Indicators Table :**
 - **IndicatorCategories:** ON DELETE CASCADE
 - **Sources:** ON DELETE RESTRICT
- **Countries Table :**
 - **Regions:** ON DELETE RESTRICT

Testing

Edit Country

Country Code

ITU

Country Code cannot be changed.

Country Name

Istanbul Technical University

Capital City

Istanbul

Region

Europe & Central Asia

Income Level

High income

Cancel

Edit Country

ISL	Iceland	Reykjavik	Europe & Central Asia	High income			
IND	India	New Delhi	South Asia	Lower middle income	Edit	Delete	View D
IDN	Indonesia	Jakarta	East Asia & Pacific	Upper middle income	Edit	Delete	View D
IRN	Iran, Islamic Rep.	Tehran	Middle East, North Africa, Afghanistan & Pakistan	Upper middle income	Edit	Delete	View D
IRQ	Iraq	Baghdad	Middle East, North Africa, Afghanistan & Pakistan	Upper middle income	Edit	Delete	View D
IRL	Ireland	Dublin	Europe & Central Asia	High income	Edit	Delete	View D
IMN	Isle of Man	Douglas	Europe & Central Asia	High income	Edit	Delete	View D
ITU	Istanbul Technical University	Istanbul	Europe & Central Asia	High income	Edit	Delete	View D
ITA	Italy	Rome	Europe & Central Asia	High income	Edit	Delete	View D
JAM	Jamaica	Kingston	Latin America & Caribbean	Upper middle income	Edit	Delete	View D

“Since this is a data-driven application with limited user interaction, our testing mainly focused on validating imported data, enforcing admin-only CRUD operations, and ensuring database integrity.”



Thank you for your attention!
